Faisal Farhan

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Dhaka, Bangladesh.

Education

M.Sc in Medical Imaging and Applications *Europe*
Erasmus Mundus Joint Master's Degree *September 2023 - June 2025*
Université de Bourgogne,France; University of Cassino, Italy; University of Girona,Spain;
Thesis: Design and Validation of Multimodal and Lightweight Handheld ST-rPPG Acquisition Systems for Physiological measurements.
GPA: 8.40/10.0;

B.Sc. in Electrical and Electronic Engineering *Dhaka, Bangladesh*
Ahsanullah University of Science & Technology *April 2016 - January 2021*
GPA: 3.936/4.0; Rank: 1/196
Undergraduate Thesis: Study of Pattern Recognition System through the development of a Raspberry Pi based speaker recognizer.

Research interests

Signal Processing, Medical Imaging, Image Processing, Machine Learning & Applied Artificial Intelligence

Work experiences

Lecturer *August 2025 - Present*
Department of Electrical and Electronic Engineering, Ahsanullah University of Science & Technology, Bangladesh.

Graduate Research Intern
Laboratoire en Imagerie et Vision Artificielle (ImVIA), Dijon, France *February 2025-June 2025*

Lecturer *June 2021 - August 2023*
Department of Electrical and Electronic Engineering, Ahsanullah University of Science & Technology, Bangladesh.

Faculty Member *May 2021 - July 2021*
School of Science, Engineering & Technology, East Delta University, Bangladesh.

Selected Publications

1. *Rizvee, Redwan Ahmed; Farrok, Omar; Hasan, Mahamudul;* ***Farhan, Faisal****; Islam, Md Hafanul; Hasan, Md Khalid; Rahman, Abidur; Islam, Maheen; Ali, Md Sawkat; Jabid, Taskeed; Rashid, Mohammad Rifat Ahmmad; Islam, Mohammad Manzurul, "Revival of Muslin by Phuti Karpas plant identification with convolution neural network", Array, Volume 27,2025, ISSN 2590-0056.* [Paper Link]
2. *Kabir, M. S. A.,* ***Farhan, F****., Siddiquee, A. A., Baroi, O. L., & Rahimi, J. (2023). "Effect of Input Channel Reduction on EEG Seizure Detection. Przeglad Elektrotechniczny", 2023(12).* [Paper Link]
3. Thahmidul I. Nafi, Erfanul Haque, **Faisal Farhan**, Asif Rahman *"High Accuracy Swin Transformers for Image-based Wafer Map Defect Detection"*, International Journal of Engineering and Manufacturing. [Paper Link]
4. **F. Farhan** and T. I. Nafi, *"ANN based approach to predict criminal trends in Bangladesh"*, 2nd International Conference on Artificial Intelligence and Signal Processing (AISP), 2022. [Paper Link]
5. T. H. Rafi, R. M. Shubair, **F. Farhan**, M. Z. Hoque and F. M. Quayyum, *"Recent Advances in Computer-Aided Medical Diagnosis Using Machine Learning Algorithms with Optimization Techniques"*, in IEEE Access. [Paper Link]
6. T. H. Rafi, **F. Farhan** (2021), *"A Performance Evaluation of Machine Learning Models on Human Activity Identification (HAI)"*,12th International Conference on Soft Computing and Pattern Recognition, Springer. [Paper Link]
7. T. H. Rafi, **F. Farhan**, Z. Hoque, F. Quayyum. (2020). *"Electroencephalogram (EEG) brainwave signal-based emotion recognition using extreme gradient boosting algorithm"*, Annals of Engineering, 1(1), 0005. [Paper Link]
8. *"Enhanced Vascular Perfusion Mapping and Heart Rate Estimation via Spatio-Temporal rPPG with Optical and Motion Compensation Techniques", Biomedical Physics and Engineering Express* [Under Review]
9. *"A Dual-Pipeline Approach for Colorectal Polyp Detection and Segmentation with Machine and Deep Learning Models", International Journal of Image, Graphics and Signal Processing (IJIGSP)* [Under Review]

SELECTED ACADEMIC PROJECTS

- Design and Validation of Multimodal and Lightweight Handheld ST-rPPG Acquisition Systems for Physiological measurements. (Masters thesis project)
- Gastrointestinal Polyp Segmentation and Classification (Advanced Image Analysis, Machine Learning and Deep Learning academic project) [pdf]
- Lung CT Volume Registration (Medical Image Registration academic project) [pdf]
- Brain MRI tissue segmentation (Medical Image Segmentation academic project) [pdf]
- Smart Chest Xray Analysis and Report Generation system (Software Engineering academic project) [github]
- Raspberry Pi based Speaker recognition system (Undergraduate thesis project)

TECHNICAL SKILLS

- Programming Languages: C++, Python, Java, Matlab, Verilog, HTML.
- Libraries: Pandas, NumPy, Matplotlib, Tkinter.
- Data analysis and visualization: Scikit learn, tensorflow, keras, pytorch, matplotlib, excel.
- Tools: Google Colab, Git, Visual Studio, Codeblocks, MATLAB, Arduino.
- Simulators: Orcad Pspice, Quartus, Proteus, Cadence.
- Design Software: Adobe Photoshop, Adobe Illustrator, Figma
- OS: Windows, Linux, Raspbian.

ACADEMIC HONORS

- Khan Bahadur Ahsanullah award (highest CGPA among all department in the graduating class)
- Erasmus Mundus Scholarship, European Union.
- Aquisis International Mobility Grant, Region Bourgogne Franche Comte (France).
- Dean's List of Honor in all four academic years.

PRESENTATIONS

- Detection of diabetic retinopathy using a fusion of textural and ridgelet features of retinal images and sequential minimal optimization classifier, *University of Burgundy (France).*
- Review on Traditional and Deep Learning Methods for Diabetic Retinopathy detection and classification, *University of Burgundy (France).* *2022*
- 2nd IEEE International Conference on Artificial Intelligence and Signal Processing (AISP).
- 12th International Conference on Soft Computing and Pattern Recognition (SOCPAR'21). *2020*

LEADERSHIP EXPERIENCES

- **Member,** Erasmus Mundus Association *2023-2025*
- **Treasurer,** AUST Career Development Club *2022-2023*
- **Chittagong Division Representative**, Bangladesh Development Project *2021*
- **Vice President**, Ahsanullah University of Science & Technology EEE Society *2020*
- **Volunteer**, Youth's Voice Bangladesh *2013-2015*